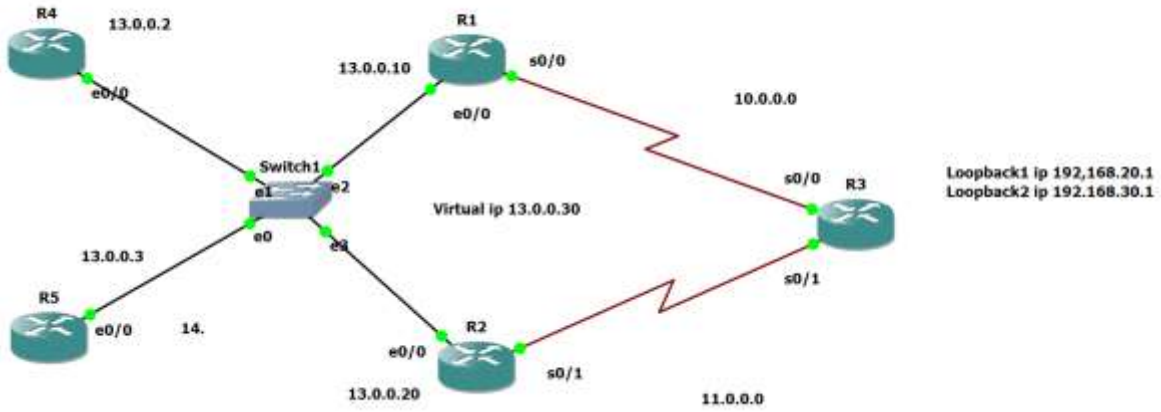
Diagram



Objective

To configure HSRP between two routers (R1 and R2) to provide gateway redundancy for the LAN. R1 will act as the Active router, and R2 will act as the Standby router, sharing a Virtual IP address

Phase 1: Make R4 and R5 act like PCs

R4 (PC Configuration):

```
R4> enable
R4# configure terminal
R4(config)# no ip routing
R4(config)# interface ethernet 0/0
R4(config-if)# ip address 13.1.1.40 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
! Point the PC to the HSRP Virtual IP as its gateway
R4(config)# ip default-gateway 13.1.1.30
```

R5 (PC Configuration):

```
R5> enable
R5# configure terminal
R5(config)# no ip routing
R5(config)# interface ethernet 0/0
R5(config-if)# ip address 13.1.1.50 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
! Point the PC to the HSRP Virtual IP as its gateway
R5(config)# ip default-gateway 13.1.1.30
```

Phase 2: Assign IPs and Run OSPF (R1, R2, R3)

R1 (Active Gateway & OSPF):

```
R1> enable
R1# configure terminal
! 1. Assign IPs
R1(config)# interface ethernet 0/0
R1(config-if)# ip address 13.1.1.10 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface serial 0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

! 2. Run OSPF Process 10, Area 1
R1(config)# router ospf 10
R1(config-router)# network 13.1.1.0 0.0.0.255 area 1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 1
```

R2 (Standby Gateway & OSPF):

```
R2> enable
R2# configure terminal
! 1. Assign IPs (Notice the .20, NOT .30)
R2(config)# interface ethernet 0/0
R2(config-if)# ip address 13.1.1.20 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# interface serial 0/1
```

```
R2(config-if)# ip address 11.0.0.1 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

! 2. Run OSPF Process 10, Area 1
R2(config)# router ospf 10
R2(config-router)# network 13.1.1.0 0.0.0.255 area 1
R2(config-router)# network 11.0.0.0 0.0.0.3 area 1
```

R3 (Core Router & OSPF):

```
R3> enable
R3# configure terminal
! 1. Assign IPs
R3(config)# interface serial 0/0
R3(config-if)# ip address 10.0.0.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface serial 0/1
R3(config-if)# ip address 11.0.0.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface loopback 1
R3(config-if)# ip address 192.168.20.1 255.255.255.0
R3(config-if)# interface loopback 2
R3(config-if)# ip address 192.168.30.1 255.255.255.0
R3(config-if)# exit

! 2. Run OSPF Process 10, Area 1
R3(config)# router ospf 10
R3(config-router)# network 10.0.0.0 0.0.0.3 area 1
R3(config-router)# network 11.0.0.0 0.0.0.3 area 1
R3(config-router)# network 192.168.20.0 0.0.0.255 area 1
R3(config-router)# network 192.168.30.0 0.0.0.255 area 1
```

Phase 3: Configure HSRP (Standby 1)

Configure R1 as Active:

```
R1(config)# interface ethernet 0/0
R1(config-if)# standby 1 ip 13.1.1.30
```

```
R1(config-if)# standby 1 priority 203
R1(config-if)# standby 1 preempt
```

Configure R2 as Standby:

```
R2(config)# interface ethernet 0/0
R2(config-if)# standby 1 ip 13.1.1.30
R2(config-if)# standby 1 priority 100
R2(config-if)# standby 1 preempt
```

Phase 4: How to Test and Verify

1. **Check HSRP:**
   On R1: `show standby brief` (Should say Active)
   On R2: `show standby brief` (Should say Standby)
2. **Check OSPF:**
   On R1: `show ip ospf neighbor` (Should see R3 as FULL)
   On R2: `show ip ospf neighbor` (Should see R3 as FULL)
3. **Test the PCs:**
   Go to **R4** (Your PC) and ping R3's loopback:
   `ping 192.168.20.1`
   If this works, your OSPF, HSRP, and PC configurations are perfect.
4. **Test Failover:**
   On R1, shut down the interface (`interface e0/0`, then `shutdown`).
   Go back to **R4** and ping `192.168.20.1` again. It should still work!
   Go to **R2** and type `show standby`. It will now say **Active**.